CITY TECHNICAL SOLUTIONS SDN. BHD.
Precision Energy Monitoring Solutions

ENERGY EYE

INSTALLATION GUIDE

Model: Energy Eye GSM 2.3v DFO 0.5
RF: NB-IoT | Voltage: 3V DC

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⚠ Safety First

Before installing or commissioning the Energy Eye device, please observe the following safety guidelines:

- **Power Safety**
  - Ensure the energy meter is accessible and safe to work on.
  - Always use insulated tools when working close to energized equipment.

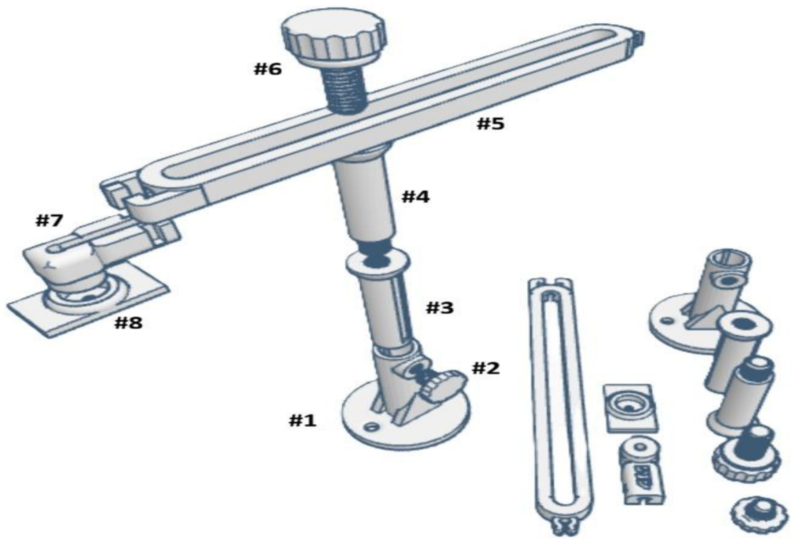
1. What's in the Box

Before starting installation, verify that all parts listed below are present and undamaged.

1.1 Energy Eye Unit

- 1 × Energy Eye main unit (black enclosure)
- 1 × Optical probe cable
- 1 × Mounting bracket set (see 1.2 for full parts list)

1.2 Bracket Parts



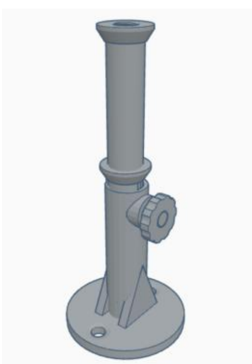
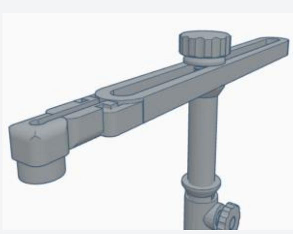
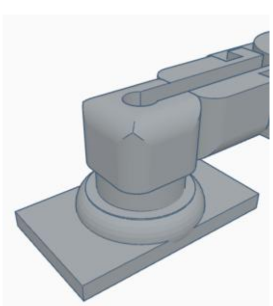
REF	QTY	PART NAME
#1	1 ×	Base Plate
#2	1 ×	Lock Knob
#3	1 ×	Vertical Support Rod
#4	1 ×	Vertical Support Rod Adaptor
#5	1 ×	Horizontal Arm
#6	1 ×	Adjustment Knob / Tightener
#7	1 ×	End Joint / Fiber Holder Mount A
#8	1 ×	Fiber Optic Holder Plate
#9	1 ×	End Joint / Fiber Holder Mount B

NOTE: If any parts are missing or damaged, contact City Technical Solutions before proceeding with installation.

2. Bracket Installation

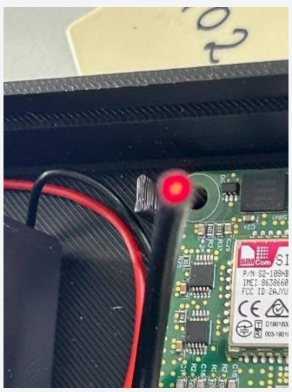
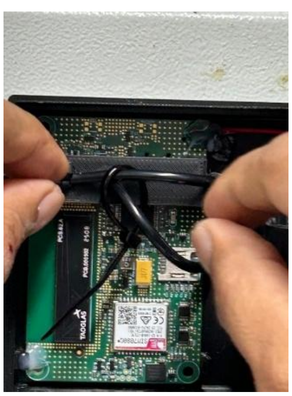
Follow the steps below to assemble and install the bracket onto the wall or panel near the energy meter.

> ⚠ **WARNING:** Ensure the energy meter is accessible before starting. Do not obstruct the kWh LED pulse indicator.

PHOTO	INSTRUCTION
	**Step 1 — Mount the Main Bracket** Secure the bracket firmly near the energy meter. 1. Use 2 screws to hang the main bracket (#1) on the wall or panel surface. 2. Ensure the bracket is level and firmly fixed before continuing.
	**Step 2 — Install the Vertical Support Rod** Attach the vertical rod for height adjustment. 1. Insert Vertical Support Rod (#3 or #4) into the main bracket. 2. Adjust the height as needed to align with the meter LED. 3. Tighten the Lock Knob (#2) securely.
	**Step 3 — Attach the Horizontal Arm** Connect the optical arm for angular alignment. 3. Fasten the Horizontal Arm (#5) using the Adjustment Knob / Tightener (#6). 4. Adjust the arm angle if necessary to point toward the meter LED.
	**Step 4 — Align the Optical Holder Plate** Position the holder plate directly over the meter LED. 5. Use the Fiber Optic Holder Plate (#8) to align with the kWh LED pulse indicator. 6. Attach the End Joint / Fiber Holder Mount A (#7) to the Holder Plate (#8). 7. Adjust height and bracket position until the probe is centred over the LED.

	**Step 5 — Connect the Optical Probe** Secure the probe as close as possible to the LED source. 8. Insert the optical probe into the Holder Plate (#8) hole. 9. Position the probe tip as close as possible to the kWh LED pulse indicator. 10. Secure the fiber optic probe with a cable tie to prevent movement during operation.

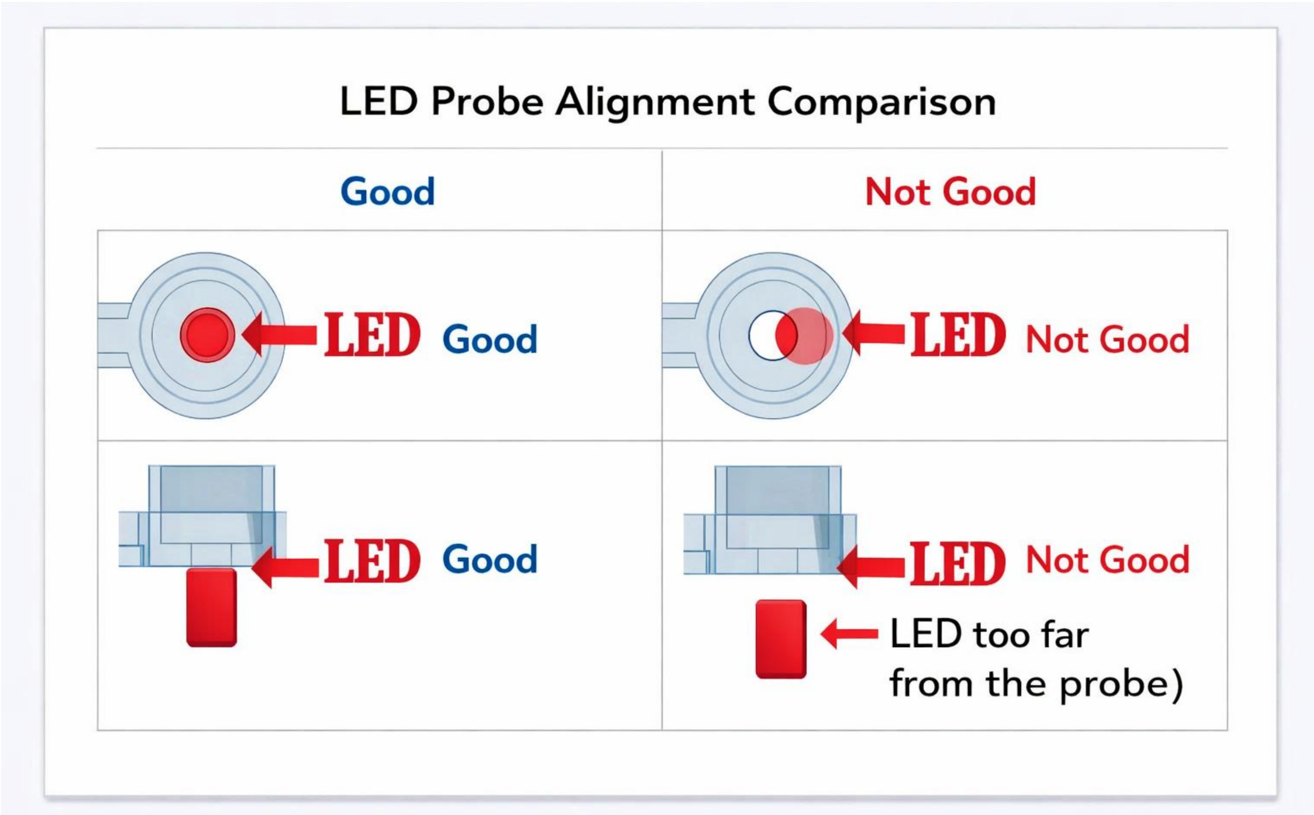
Optical Alignment Check

PHOTO	INSTRUCTION
	**Verify Optical Alignment** 11. Remove End Joint B (#9) from the Fiber Optic Holder Plate. 12. Look through the optical path to verify the probe is centred over the kWh meter LED (Blinking). 13. If the LED is not centred, adjust the height adjuster and reposition the optical bracket until the probe sits directly over the LED pulse indicator.
	**Reinsert and Secure End Joint B** 14. Once LED alignment is confirmed, reinsert End Joint B (#9) into the Fiber Optic Holder Plate. 15. Ensure it is seated firmly in the hole and positioned close to the Energy Eye lux sensor. 16. Use a cable tie to secure the fiber optic probe and prevent movement or misalignment during operation.

3. Optical Probe Alignment

Correct alignment of the optical probe to the kWh LED is critical for accurate pulse counting. Use the reference table below as a guide.

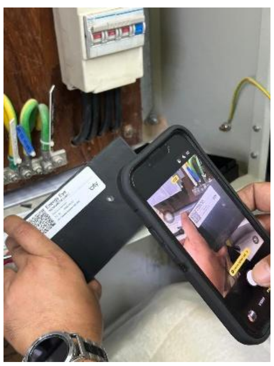
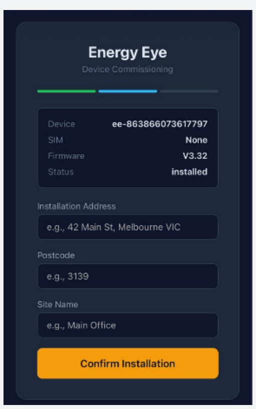
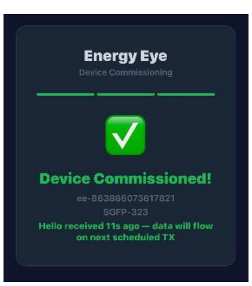
3.1 Alignment Status Reference — End Joint A (#7)



STATUS	ALIGNMENT	DESCRIPTION & ACTION
☑ GOOD	Probe fully aligned with LED	The optical probe is directly facing the kWh LED. The probe lens fully covers the LED. This will produce accurate pulse readings.
✕ NOT GOOD	Probe misaligned	The probe is not directly facing the LED or is too far from it. Adjust the height adjuster and optical bracket until the probe is fully aligned.
✕ NOT GOOD	LED too far from probe	The LED pulse indicator is too far from the optical probe tip. Slide the probe closer using the height extender until the probe tip is as close as possible to the LED.

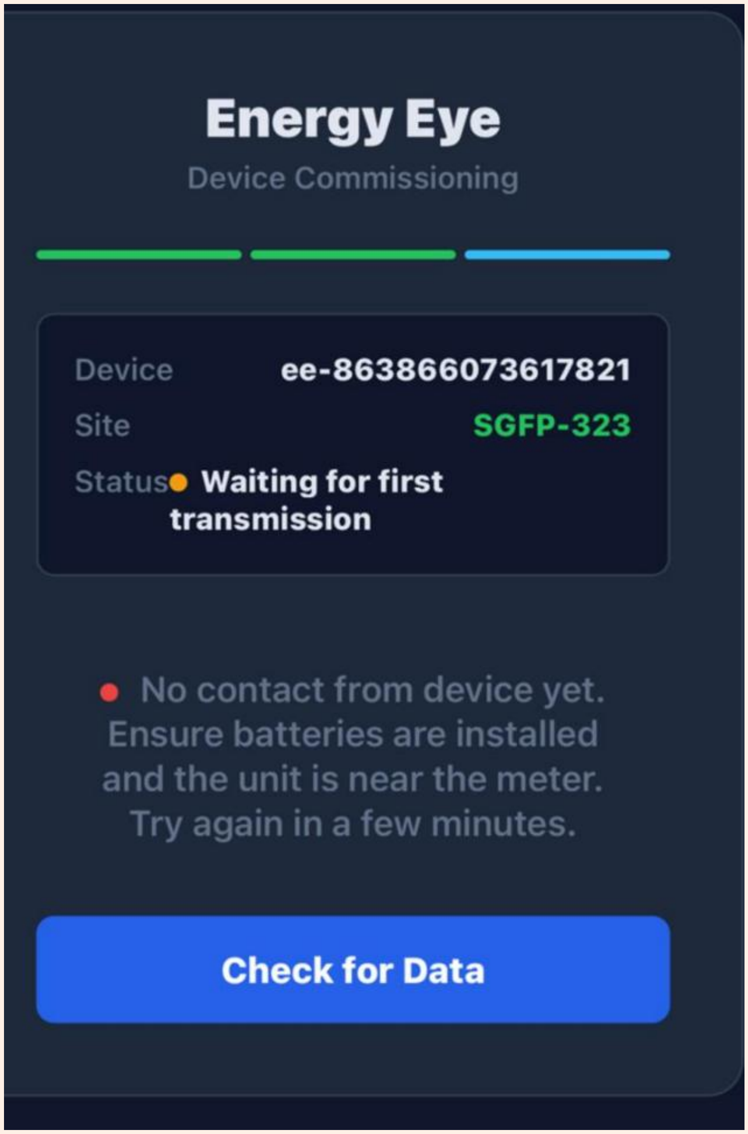
4. Energy Eye Unit Setup

After the bracket is installed and the probe is aligned, set up the Energy Eye main unit by following the steps below.

<table>
<tr><th>PHOTO</th><th>INSTRUCTION</th></tr>
<tr><td>
</td><td>Insert Batteries
Insert 2 × D-size batteries into the battery compartment, ensuring the correct polarity (+/−) as marked inside the compartment.</td></tr>
<tr><td></td><td>Scan the Barcode
Use the scanner to scan the barcode of the device for registration and verification.</td></tr>
<tr><td>
</td><td>Complete Installation Fields
17. Enter the installation address.
18. Enter the postcode.
19. Enter the site name.
20. Press "Confirm Installation" once all fields are completed.</td></tr>
<tr><td>
</td><td>Verify Confirmation Message
Wait for the confirmation message to appear on screen and verify that the registration has been accepted successfully.</td></tr>
</table>

Troubleshooting

If the device is not transmitting data after installation, carry out the following checks:



- Confirm the battery is inserted with the correct polarity (+/−).
- Ensure all fields on the commissioning page are completed (installation address, postcode, site name).
- Try relocating the device to a higher position for better network signal performance.
- Verify optical probe alignment — the probe must be centred over the kWh LED pulse indicator.

📞 **CONTACT:** If all the above steps are completed and the issue persists, please contact your supplier for technical support.

5. Post-Installation Verification

Before leaving the installation site, confirm the following checklist is complete:

- Optical probe is aligned and as close as possible to the kWh LED pulse indicator.
- All 2 × D batteries are correctly inserted with the correct polarity.
- The Energy Eye unit is securely mounted and the bracket is tightened.
- The Energy Eye device is transmitting data — check the backend dashboard for the first data packet within 10 minutes of installation.

📌 **NOTE:** Contact City Technical Solutions if no data appears on the backend within 30 minutes of installation.

6. Technical Specifications

SPECIFICATION	VALUE
Product Name	Energy Eye
Model	Energy Eye GSM 2.3v DFO 0.5
RF Technology	GSM / LTE / NB-IoT
Voltage	3V DC (2 × D 1.5V battery banks)
Power Supply	2 × D batteries
Battery Life	Up to 10 years (typical)
Pulse Detection	Optical (kWh LED pulse)
Pulse Rate	pulses/kWh — check meter label
Operating Temperature	0°C to +55°C
IP Rating	IP54
Dimensions	120 × 90 × 40 mm
Weight	To be confirmed
Warranty	12 months

City Technical Solutions Sdn. Bhd. — *Changes to specifications may occur without prior notice.*